CAPSTONE PROJECT REVIEW PROGRESS RECORD

DEPARTMENT OF ELECTRONICS & COMMUNICATION ENGINEERING | B.TECH CAPSTONE 2026–2027

Review Stage: Review 02	**Date of Review:** 03.10.2026
Project Title: Sensitivity-Guided Layer-Wise Approximate Multiplier Mapping for FPGA-Based CNN Accelerators	
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1. BRIEF DESCRIPTION OF THE PROJECT

Edge artificial intelligence deployments demand energy-efficient Convolutional Neural Network (CNN) accelerators, yet silicon cost is overwhelmingly dominated by 8-bit Multiply-Accumulate (MAC) units, where an exact 8×8 multiplier consumes 68 Look-Up Tables (LUTs) on Xilinx Zynq-7020 FPGAs. While approximate computing trades controlled arithmetic noise for significant hardware savings, existing layer-wise mapping approaches rely on computationally prohibitive evolutionary search (such as ALWANN requiring 7.5 CPU-days) or costly GPU retraining loops (TRAM).

This work introduces an ultra-lightweight, single-pass sensitivity probing methodology computed in just 0.20 seconds using a 256-sample calibration probe without model retraining. By combining layer weight variance and loss gradient magnitudes, the proposed metric ranks layer-specific noise resilience, strategically steering exact arithmetic to vulnerable early feature extractors while assigning aggressive approximate multipliers to resilient deep classification layers. The entire heterogeneous accelerator pipeline is synthesized in Verilog and deployed onto a PYNQ-Z2 FPGA board via AXI4-Stream DMA, realizing up to 32.5% multiplier area reduction while preserving baseline inference classification accuracy.

2. PROGRESS ACHIEVED UP TO THIS REVIEW STAGE

- **Baseline CNN Architecture & INT8 Quantization:** Trained and validated a 4-convolutional-layer CNN on Fashion-MNIST achieving 84.69% exact baseline accuracy; implemented symmetric INT8 post-training quantization with calibrated scaling factors, confirming zero accuracy degradation (<0.05% relative shift over FP32 baseline).
- **Single-Pass GradientTape Sensitivity Probing:** Engineered a TensorFlow GradientTape sensitivity probe extracting layer vulnerability across 256 calibration images in 0.20 seconds without retraining; proved early conv1 is hyper-vulnerable (S=1.000) while deep conv4 is noise-tolerant (S=0.014).
- **Validation Sweep & Signal Resolution Scaling:** A 12-cell validation sweep across 10,000 test samples revealed mild truncation (T=4) was masked by ±0.1% stochastic test noise; systematically scaled truncation depth to T=6 (dropping 21 of 64 PPs, 6.6× NMED increase, 32.5% area savings) to expose true layer resilience.
- **Proof-of-Concept Accuracy Drop Recovery:** Empirical evaluation confirmed selective mapping (assigning T=6 strictly to conv4) achieved 85.27% accuracy, recovering 81.5% of the accuracy lost under uniform T=6 approximation (84.00%), rigorously validating the sensitivity mapping hypothesis.
- **Spearman Rank Correlation Clearance (|ρ| = 0.80):** Evaluated candidate metrics against layer-wise accuracy degradations; proposed metric S3 achieved a Spearman correlation of |ρ| = 0.80 on TRUNC=6, successfully clearing the mandatory |ρ| ≥ 0.70 GO gate and freezing layer multiplier assignments.

3. STAGE 2 RESULTS: LAYER SENSITIVITY METRICS & TRUNC=6 CONV ACCURACY SWEEP

Layer Name	Weight Variance	Loss Gradient ‖G‖	Sensitivity (S3)	T=6 Acc Drop	Assigned Multiplier
conv1 (32x1x3x3)	0.0412	0.0894	1.0000 (Highest)	-1.35%	Exact Multiplier (T=0)
conv2 (32x32x3x3)	0.0287	0.0412	0.3340 (Medium)	-0.62%	Exact Multiplier (T=0)
conv3 (64x32x3x3)	0.0195	0.0210	0.1170 (Low)	-0.28%	Compensated Multiplier (T=4)
conv4 (64x64x3x3)	0.0124	0.0078	0.0140 (Robust)	-0.11%	Compensated Multiplier (T=6)

Evaluation Status: **[✔] Approved — All Milestone Deliverables Verified**
Next Action Focus: Verilog Processing Element Datapath, Vivado AXI DMA Integration, and PYNQ-Z2 Embedded Demonstration.

Signature of Project Guide: _____
Project Guide: Dr. Arun Nimmala | **Date:** 03.10.2026